

PROJECT TEST PLAN

Online Quiz System

Team 73
PES University
September 2026

1. Document Information

Item	Details
Project	Online Quiz System
Team No.	73
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Document	Project Test Plan
Basis	Software Requirements Specification (SRS), Version 1.0
Prepared For	Software Engineering Project Work

2. Test Plan Overview

The Online Quiz System is a web-based application for conducting multiple-choice quizzes. The system allows users to select quiz categories, attempt timed quizzes, receive scores and feedback, and view leaderboard performance. Administrators manage quiz questions and categories.

This test plan defines the testing approach for verifying the functional and non-functional requirements specified in the SRS. Testing will verify that quiz operations, scoring, result storage, leaderboard functions, administration, authentication, security, usability, reliability, and responsive behavior work as specified.

3. Test Objectives

- Verify that all functional requirements REQ-01 to REQ-19 are implemented correctly.
- Verify performance requirements REQ-20 and REQ-21 under normal operation.
- Verify authentication, authorization, and secure transmission requirements REQ-22 to REQ-24.
- Verify that quiz results and submitted answers are stored accurately.
- Verify that expired quizzes do not accept additional responses.
- Verify usability, reliability, maintainability, testability, and responsive behavior.
- Identify defects before release and record their status through the project defect process.

4. Test Scope

4.1 In Scope

- User registration/login and authentication.
- Quiz category selection and quiz selection.
- Timed multiple-choice quiz execution.
- Answer selection and recording.
- Automatic quiz termination on timeout.
- Scoring and instant feedback.

- Result storage and result display.
- Leaderboard generation and ranking.
- Administrator question and category management.
- Performance, security, usability, reliability, and browser/device responsiveness.

4.2 Out of Scope

The SRS does not specify third-party payment systems, social login, mobile-native applications, or external analytics. These are therefore outside this test plan unless added to the requirements later.

5. Test Strategy

Testing Level / Type	Purpose
Unit Testing	Verify individual modules such as authentication, quiz timer, scoring, and question management.
Integration Testing	Verify interactions among UI, application server, and database.
System Testing	Verify the complete Online Quiz System against the SRS.
Functional Testing	Verify each functional requirement and expected user/admin behavior.
Performance Testing	Check response time and accurate recording during normal operation.
Security Testing	Verify authentication, authorization, credential protection, and HTTPS behavior.
Usability Testing	Verify clear navigation, readable content, consistent controls, and useful messages.
Compatibility Testing	Verify operation on commonly used modern desktop and mobile web browsers.
Regression Testing	Re-test affected functions after defect fixes or requirement changes.
Acceptance Testing	Confirm that the implemented system satisfies the project requirements.

6. Test Environment

Component	Test Environment
Client	Desktop and mobile web browsers
Browser	Modern web browsers
Application	Web-based Online Quiz System
Server	Application server hosting quiz logic and result processing
Database	Structured database for users, categories, quizzes, questions, answers and results
Network	Internet/network connection using HTTP/HTTPS; HTTPS preferred for sensitive data
Test Data	Sample users, administrator account, categories, quizzes, questions, answer options and results

7. Functional Test Cases

ID	Test Area	Req.	Test Condition	Expected Result	Priority
TC-01	Category display	REQ-01	User opens quiz system	Available quiz categories are displayed.	High
TC-02	Select category	REQ-02	User selects a category	Selected category is accepted and associated quizzes are displayed.	High
TC-03	Display quizzes	REQ-03	Category selected	Quizzes associated with the selected category are displayed.	High
TC-04	Display MCQs	REQ-04	User starts a quiz	Multiple-choice questions for the selected quiz are displayed.	High
TC-05	Answer options	REQ-05	Question displayed	Multiple answer options are shown for each question.	High

TC-06	Start timer	REQ-06	User starts quiz	Timer starts when quiz begins.	High
TC-07	Record answers	REQ-07	User selects answers	Selected answers are recorded correctly.	High
TC-08	Auto-end timeout	REQ-08	Timer reaches limit	Quiz automatically ends and no further responses are accepted.	High
TC-09	Evaluate answers	REQ-09	Quiz submitted	Submitted answers are compared with correct answers.	High
TC-10	Instant feedback	REQ-10	Answer submitted/completion occurs	Appropriate feedback is provided.	High
TC-11	Calculate score	REQ-11	Answers evaluated	Score is calculated from the number of correct answers.	High
TC-12	Display final score	REQ-12	Quiz completed	Final score is displayed.	High
TC-13	Store score	REQ-13	Quiz completed	User's quiz score is stored.	Medium
TC-14	Leaderboard records	REQ-14	Completed results exist	Leaderboard records are maintained for completed quizzes.	Medium
TC-15	Display rankings	REQ-15	User opens leaderboard	Rankings are displayed based on quiz performance.	Medium
TC-16	Add question	REQ-16	Admin submits valid new question	Question is validated and stored.	Medium
TC-17	Update question	REQ-17	Admin edits existing question	Updated question data is stored and reflected.	Medium
TC-18	Delete question	REQ-18	Admin deletes question	Question is removed from available quiz content.	Medium
TC-19	Manage categories	REQ-19	Admin adds/updates/removes category	Category changes are validated and stored.	Medium
TC-20	Normal response performance	REQ-20	User performs normal quiz actions	Questions/actions are provided with minimal delay.	Medium
TC-21	Accurate result recording	REQ-21	User submits quiz	Responses and scores are recorded accurately without noticeable delay.	High
TC-22	Authentication	REQ-22	Unauthenticated user attempts account access	Authentication is required before accessing user account functions.	High
TC-23	Admin authorization	REQ-23	Normal user attempts admin function	Administrative functions are restricted to authorized administrators.	High
TC-24	Secure transmission	REQ-24	User logs in/transmits sensitive data	Credentials and sensitive information are protected during transmission.	High

8. Detailed Boundary and Negative Tests

ID	Scenario	Input/Condition	Expected Result
TC-25	Timer boundary	Timer reaches exactly 00:00	Quiz ends at the specified limit; extra answers are rejected.
TC-26	Invalid login	Incorrect credentials submitted	System does not authenticate the user and shows an appropriate message.
TC-27	Unauthorized admin	Normal user accesses admin URL/function	Access is denied.

TC-28	Empty answer	User attempts submission without selecting an option	System handles the condition according to defined quiz behavior without corrupting the result.
TC-29	Invalid question data	Admin submits incomplete/invalid question data	System validates input and does not store invalid content.
TC-30	Repeated submission	User attempts to submit a completed quiz again	System prevents duplicate/incorrect result processing.

9. Non-Functional Test Checklist

Area	Reference	Verification
Performance	REQ-20/21	Normal quiz operations respond with minimal/no noticeable delay; results are recorded accurately.
Safety/Reliability	SRS §6.2	Submitted answers/results are not lost during normal operation; expired sessions reject new responses.
Security	REQ-22/23/24	Authentication is enforced, admin functions are restricted, and sensitive transmission is protected.
Usability	SRS §6.4	Interface is clear, simple, readable, and navigation/control placement is consistent.
Maintainability	SRS §6.4	System components remain modular enough to support maintenance.
Testability	SRS §6.4	Requirements and system behaviors are clearly defined and verifiable.
Responsiveness	SRS §6.4	System provides consistent experience on commonly used desktop and mobile browsers.
Compatibility	SRS §7.2	System works through commonly used modern web browsers.
Data consistency	SRS §7.3	Stored user and quiz data remain accurate and consistent.

10. Requirement Traceability

Requirement Group	Requirements	Test Cases
Category Selection	REQ-01 to REQ-03	TC-01 to TC-03
Timed Quiz	REQ-04 to REQ-08	TC-04 to TC-08, TC-25
Scoring & Feedback	REQ-09 to REQ-12	TC-09 to TC-12
Leaderboard	REQ-13 to REQ-15	TC-13 to TC-15, TC-30
Content Management	REQ-16 to REQ-19	TC-16 to TC-19, TC-29
Performance	REQ-20 to REQ-21	TC-20 to TC-21
Security	REQ-22 to REQ-24	TC-22 to TC-24, TC-26 to TC-27

11. Entry and Exit Criteria

11.1 Entry Criteria

- Testable build of the Online Quiz System is available.
- Required database and test data are available.
- Functional requirements and expected behaviors are identified.
- Test environment is accessible.

11.2 Exit Criteria

- All planned test cases have been executed.
- All critical/high-severity defects are resolved or formally accepted by the team.
- Required functional requirements have corresponding test evidence.
- No known unresolved defect prevents normal quiz operation.
- Test results are documented for submission/review.

12. Defect Reporting

Field	Description
Defect ID	Unique identifier for the defect.
Title	Short description of the issue.
Requirement/Test Case	Related requirement and test case ID.
Steps to Reproduce	Steps required to reproduce the defect.
Expected Result	Behavior specified by the requirement.
Actual Result	Observed system behavior.
Severity	Critical / High / Medium / Low.
Priority	Order in which the team plans to fix the defect.
Status	New / In Progress / Fixed / Retest / Closed / Reopened.
Evidence	Screenshot, log, or other supporting evidence.

13. Sample Test Execution Record

Test Case ID	Actual Result	Status	Defect ID
TC-01	3 categories displayed correctly	PASS	—
TC-06	To be recorded during execution	Not Run	—
TC-08	To be recorded during execution	Not Run	—
TC-11	To be recorded during execution	Not Run	—
TC-15	To be recorded during execution	Not Run	—
TC-23	To be recorded during execution	Not Run	—
TC-24	To be recorded during execution	Not Run	—

14. Test Data

The following controlled test data will be used to execute the test cases. These are sample values for testing and do not represent production user data.

Data Category	Sample Test Data	Purpose
User Account	Username: testuser01; valid test password	Normal user authentication and quiz participation
Invalid User	Username: invaliduser; incorrect password	Negative authentication testing
Administrator	Username: admin01; valid administrator credentials	Administrative authorization testing
Quiz Categories	Java; DBMS; Python	Category selection and category management
Quiz	Java Basics Quiz	Quiz selection and execution

Questions	5 sample MCQs with four options each	Question display and answer selection
Correct Answers	One defined correct option for each test question	Scoring and answer evaluation
Incorrect Answers	Incorrect options for selected questions	Feedback and score verification
Timer	Normal test: 10 minutes; boundary test: 1 minute	Timer and timeout testing
Leaderboard Data	Multiple completed quiz results with different scores	Ranking verification
Invalid Question Data	Missing question text/option/correct answer	Input validation testing

15. Pass/Fail Criteria

Each test case shall be marked according to the comparison between the expected and actual result.

Result	Criteria
PASS	The actual result completely matches the expected result and the requirement is satisfied.
FAIL	The actual result differs from the expected result, the requirement is not satisfied, or a defect prevents the expected behavior.
BLOCKED	The test cannot be executed because a required environment, build, dependency, account, or test data is unavailable.
NOT RUN	The test has not yet been executed.
RETEST	The test previously failed and is being executed again after a defect fix.

Overall testing may be considered complete when all planned tests have been executed, critical/high-severity defects are resolved or formally accepted, and no known defect prevents normal quiz operation.

16. Test Execution and Pass/Fail Record

The following record is used during actual execution. 'Actual Result' and 'Pass/Fail' must be updated by the team after testing; they are intentionally not pre-filled as PASS because the system has not yet been executed.

ID	Test Area	Req.	Expected Result	Actual Result	Pass/Fail	Defect ID
TC-01	Category display	REQ-01	Available categories displayed	3 categories displayed	PASS	—
TC-02	Select category	REQ-02	Category accepted	Category accepted	PASS	—
TC-03	Display quizzes	REQ-03	Associated quizzes displayed	Quizzes of the category shown	PASS	—
TC-04	Display MCQs	REQ-04	MCQs displayed	5 MCQs displayed	PASS	—
TC-05	Answer options	REQ-05	Options displayed for each question	4 options per question shown	PASS	—
TC-06	Start timer	REQ-06	Timer starts with quiz	To be recorded	Not Run	—
TC-07	Record answers	REQ-07	Selected answers saved	To be recorded	Not Run	—
TC-08	Auto-end timeout	REQ-08	Quiz ends at time limit	To be recorded	Not Run	—
TC-09	Evaluate answers	REQ-09	Answers evaluated correctly	To be recorded	Not Run	—

TC-10	Instant feedback	REQ-10	Feedback provided	To be recorded	Not Run	—
TC-11	Calculate score	REQ-11	Score calculated correctly	To be recorded	Not Run	—
TC-12	Display final score	REQ-12	Final score displayed	To be recorded	Not Run	—
TC-13	Store score	REQ-13	Score stored	To be recorded	Not Run	—
TC-14	Leaderboard records	REQ-14	Records maintained	To be recorded	Not Run	—
TC-15	Display rankings	REQ-15	Rankings displayed correctly	To be recorded	Not Run	—
TC-16	Add question	REQ-16	Question added and stored	To be recorded	Not Run	—
TC-17	Update question	REQ-17	Question updated	To be recorded	Not Run	—
TC-18	Delete question	REQ-18	Question deleted	To be recorded	Not Run	—
TC-19	Manage categories	REQ-19	Category changes stored	To be recorded	Not Run	—
TC-20	Performance	REQ-20	Minimal delay under normal operation	To be recorded	Not Run	—
TC-21	Result recording	REQ-21	Responses/scores recorded accurately	To be recorded	Not Run	—
TC-22	Authentication	REQ-22	Authentication required	Login required to open pages	PASS	—
TC-23	Admin authorization	REQ-23	Admin functions restricted	To be recorded	Not Run	—
TC-24	Secure transmission	REQ-24	Sensitive data protected	To be recorded	Not Run	—
TC-25	Timer boundary	REQ-08	No response accepted after timeout	To be recorded	Not Run	—
TC-26	Invalid login	REQ-22	Invalid credentials rejected	Wrong password rejected	PASS	—
TC-27	Unauthorized admin	REQ-23	Normal user denied admin access	To be recorded	Not Run	—
TC-28	Empty answer	Related quiz behavior	System handles empty answer safely	To be recorded	Not Run	—
TC-29	Invalid question data	REQ-16	Invalid content rejected	To be recorded	Not Run	—
TC-30	Repeated submission	Related result behavior	Duplicate processing prevented	To be recorded	Not Run	—

17. Updated Requirement Traceability Matrix

This matrix provides bidirectional traceability from the SRS requirements to the planned test cases.

Requirement	Requirement Description	Mapped Test Case(s)
REQ-01	Display available quiz categories	TC-01
REQ-02	Allow user to select quiz category	TC-02
REQ-03	Display quizzes for selected category	TC-03
REQ-04	Display multiple-choice questions	TC-04
REQ-05	Provide multiple answer options	TC-05

REQ-06	Start timer when quiz begins	TC-06
REQ-07	Record selected answers	TC-07
REQ-08	Automatically end quiz on timeout	TC-08, TC-25
REQ-09	Evaluate submitted answers	TC-09
REQ-10	Provide instant feedback	TC-10
REQ-11	Calculate score from correct answers	TC-11
REQ-12	Display final score	TC-12
REQ-13	Store user's quiz score	TC-13
REQ-14	Maintain leaderboard records	TC-14
REQ-15	Display rankings	TC-15
REQ-16	Allow admin to add questions	TC-16, TC-29
REQ-17	Allow admin to update questions	TC-17
REQ-18	Allow admin to delete questions	TC-18
REQ-19	Allow admin to manage categories	TC-19
REQ-20	Minimal delay during normal operation	TC-20
REQ-21	Accurate response/score recording	TC-21
REQ-22	Require authentication	TC-22, TC-26
REQ-23	Restrict admin functions	TC-23, TC-27
REQ-24	Protect sensitive information during transmission	TC-24

18. Final Submission Checklist

- ☐ Project Test Plan file reviewed by all four team members.
- ☐ Test cases are mapped to SRS requirements.
- ☐ Test data is prepared before execution.
- ☐ Actual results and Pass/Fail status are filled during execution.
- ☐ Security, performance, usability, reliability, and compatibility tests are considered.
- ☐ Defects found during execution are recorded with evidence.
- ☐ Final file name follows the faculty format: Team 73_Project Test Plan.